

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Part 2 - Solving $\phi_\omega(x, t)$

- We are now interested in solving the following 1D problem with a ZRP

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases} \quad (34)$$

where

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \varkappa \delta(x), \quad \hat{V}_{\text{ext}}(t) = F(t)x, \quad (35)$$

$$\phi_\omega(x, 0) = \frac{-i\varkappa^{3/2}}{\omega} \text{sgn}(x) \left(e^{-\varkappa|x|} - e^{-\varkappa(\omega)|x|} \right), \quad \psi(x, 0) = \varkappa^{1/2} e^{-\varkappa|x|}, \quad (36)$$

with

$$\hat{p} = -id/dx, \quad \text{and} \quad \varkappa(\omega) = \sqrt{\varkappa^2 + 2\omega}. \quad (37)$$

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Solutions to the homogeneous and inhomogeneous equations

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases}$$

$$\psi(x, t) = e^{-i \int_{t_0}^t (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} \psi(x, 0), \quad (38)$$

$$\phi_\omega(x, t) = e^{-i \int_{t_0}^t (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} \phi_\omega(x, 0) - i e^{-i \int_{t_0}^t (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} \int_{t_0}^t e^{i \int_{t_0}^s (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} e^{i\omega s} \hat{p} \psi(x, s) ds \quad (39)$$

$$\begin{aligned} \psi(x, t) &= U(t, t_0) \psi(x, 0), & U(t, t_0) &= e^{-i \int_{t_0}^t (\hat{H}_a + \hat{V}_{\text{ext}}(\tau)) d\tau} \\ \phi_\omega(x, t) &= U(t, t_0) \phi_\omega(x, 0) - U(t, t_0) \int_{t_0}^t U(s, t_0)^{-1} e^{i\omega s} \frac{\partial}{\partial x} \psi(x, s) ds \end{aligned} \quad (40)$$

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Time propagation

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases}$$

$$\psi(x, t + \Delta t) = U(t + \Delta t, t) \psi(x, t),$$

$$\phi_\omega(x, t + \Delta t) = U(t + \Delta t, t) \phi_\omega(x, t) - \int_t^{t+\Delta t} U(t + \Delta t, s) e^{i\omega s} \frac{\partial}{\partial x} \psi(x, s) ds$$

■ From the trapezoidal rule applied to the integral, we get

$$\psi(x, t + \Delta t) = U(\Delta t) \psi(x, t), \quad U(t + \Delta t, t) = U(\Delta t) \quad (41)$$

$$\phi_\omega(x, t + \Delta t) = U(\Delta t) \phi_\omega(x, t) - \frac{\Delta t}{2} \left(e^{i\omega(t+\Delta t)} \frac{\partial}{\partial x} \psi(x, t + \Delta t) + U(\Delta t) e^{i\omega t} \frac{\partial}{\partial x} \psi(x, t) \right) \quad (42)$$

$$= U(\Delta t) \left(\phi_\omega(x, t) - \frac{\Delta t}{2} e^{i\omega t} \frac{\partial}{\partial x} \psi(x, t) \right) - \frac{\Delta t}{2} e^{i\omega(t+\Delta t)} \frac{\partial}{\partial x} \psi(x, t + \Delta t) \quad (43)$$

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Projections

$$p_{k0} = \int \psi_k^{(-)*}(x) \hat{p} \psi_0(x) dx = \frac{2k\mathcal{K}^{3/2}}{\mathcal{K}^2 + k^2} \quad (44)$$

$$p_{kk'} = \int \psi_k^{(-)*}(x) \hat{p} \psi_{k'}^{(-)}(x) dx = 2\pi k \delta(k - k') - \frac{2\mathcal{K}}{k^2 - k'^2 + i0} \left(\frac{k'|k|}{|k| - i\mathcal{K}} - \frac{k|k'|}{|k'| + i\mathcal{K}} \right) \quad (45)$$

$$b_{0\omega}(T) = e^{iE_0 T} \int \psi_0(x) \phi_\omega(x, T) dx, \quad (46)$$

$$b_{k\omega}(T) = e^{iE_k T} \int \psi_k^{(-)*}(x) \phi_\omega(x, T) dx, \quad (47)$$

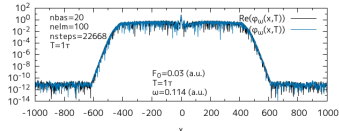
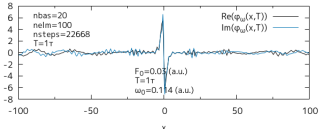
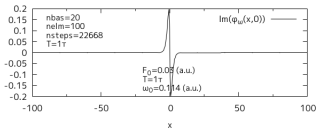
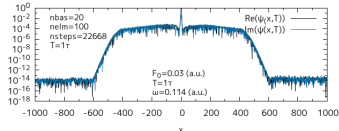
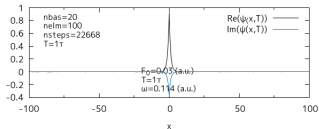
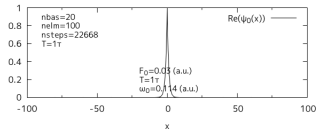
$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi} \quad (48)$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi} \quad (49)$$

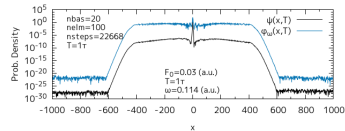
$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi} \quad (50)$$

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The wavefunction $\phi_\omega(x, t)$ ($\omega = 1$)

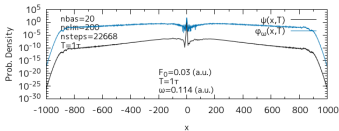
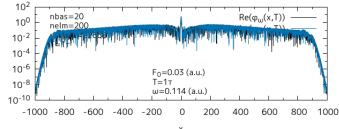
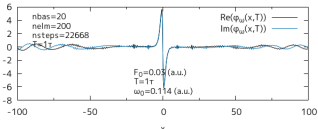
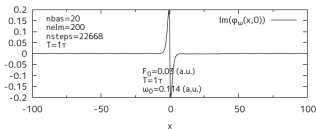
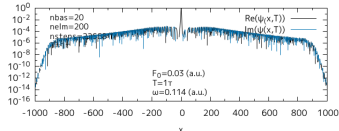
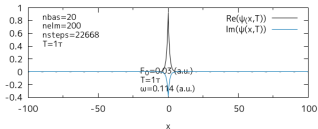
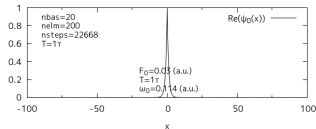


nbas = 20
nelm = 100
nsteps = 22668
xmax = 2000
T = 1τ



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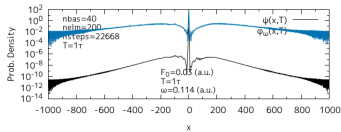
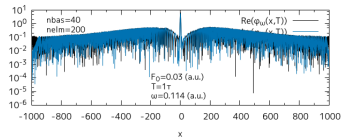
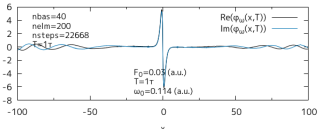
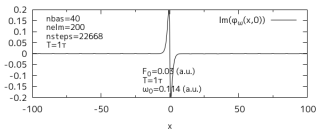
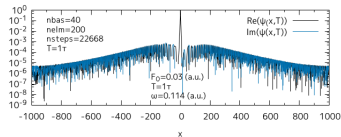
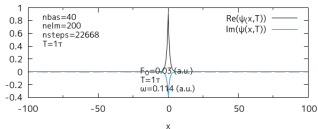
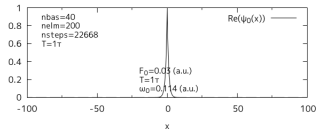
The wavefunction $\phi_\omega(x, t)$ ($\omega = 1$)



nbas = 20
nelm = 200
nsteps = 22668
xmax = 2000
T = 1τ

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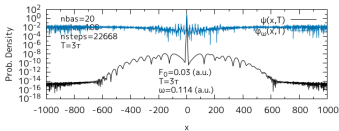
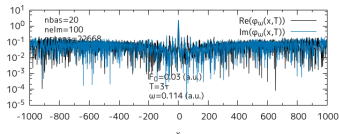
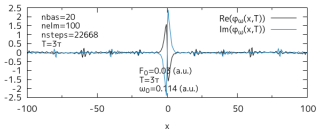
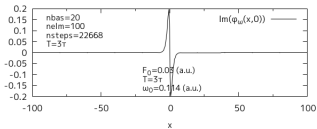
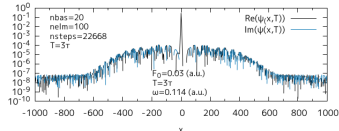
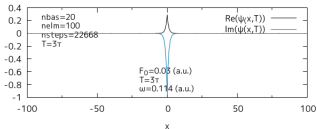
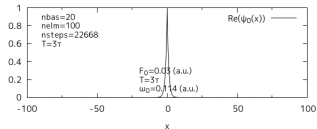
The wavefunction $\phi_\omega(x, t)$ ($\omega = 1$)



nbas = 40
nelm = 200
nsteps = 22668
xmax = 2000
T = 1τ

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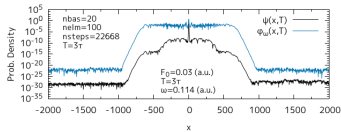
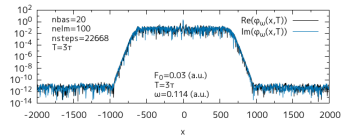
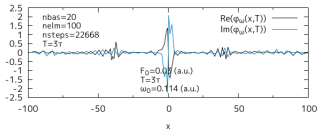
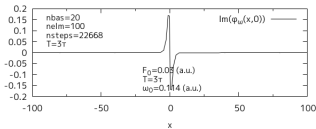
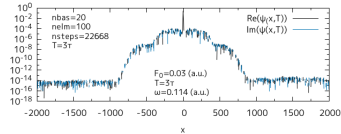
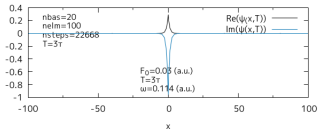
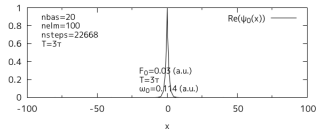
The wavefunction $\phi_\omega(x, t)$ ($\omega = 1$)



nbas = 20
nelm = 100
nsteps = 22668
xmax = 2000
T = 3τ

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The wavefunction $\phi_\omega(x, t)$ ($\omega = 1$)



nbas = 20
nelm = 100
nsteps = 22668
xmax = 4000
 $T = 3\tau$